
Endogenous portfolios and risk premia

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Portfolios in heterogeneous-agent macro

- ❖ **So far:** single asset models (from the perspective of households)
- ❖ In models with several assets available, used “mutual fund trick”
 - ❖ see stocks + bonds model, but could also do this with nominal and real bonds, short and long term bonds, home and foreign bonds...
 - ❖ from this we got no-arbitrage conditions + initial r_0^p
- ❖ **Now:** what if we let households trade these different assets directly?
 - ❖ MIT shock world: portfolios are **indeterminate**
 - ❖ can feed in distribution of portfolios from the data
 - ❖ Expected shocks world: portfolios are **determinate!**

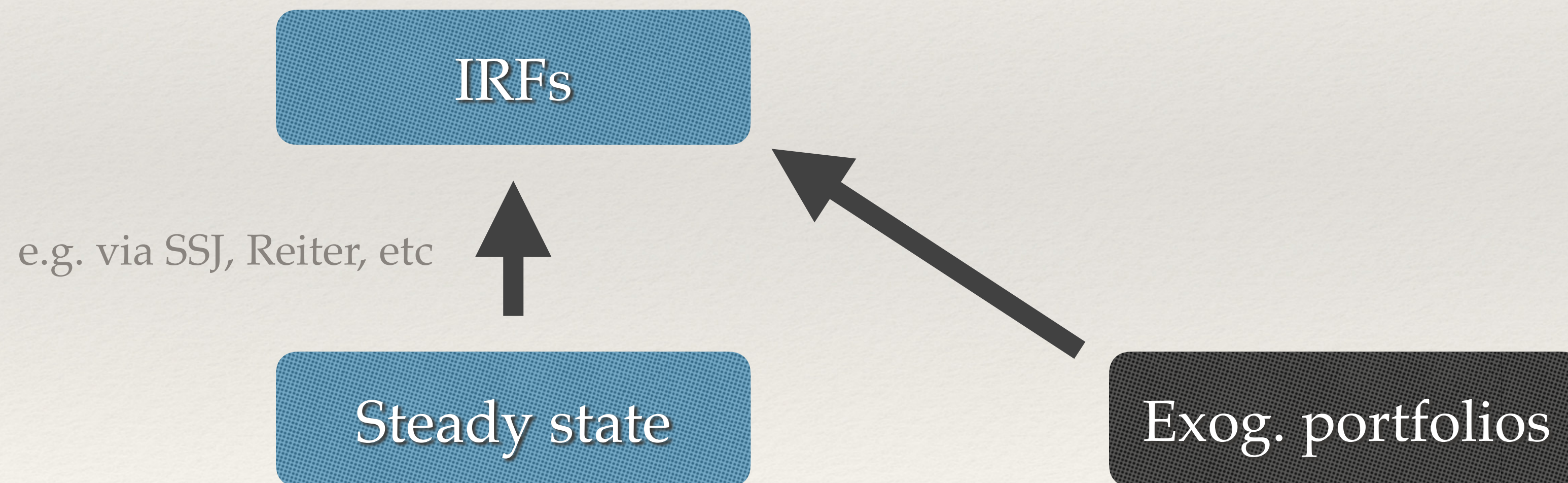
What does second order perturbation give us?

Orders in aggregate risk

2nd order

1st order

0th order



What does second order perturbation give us?

Orders in aggregate risk

2nd order

Steady state
risk premium

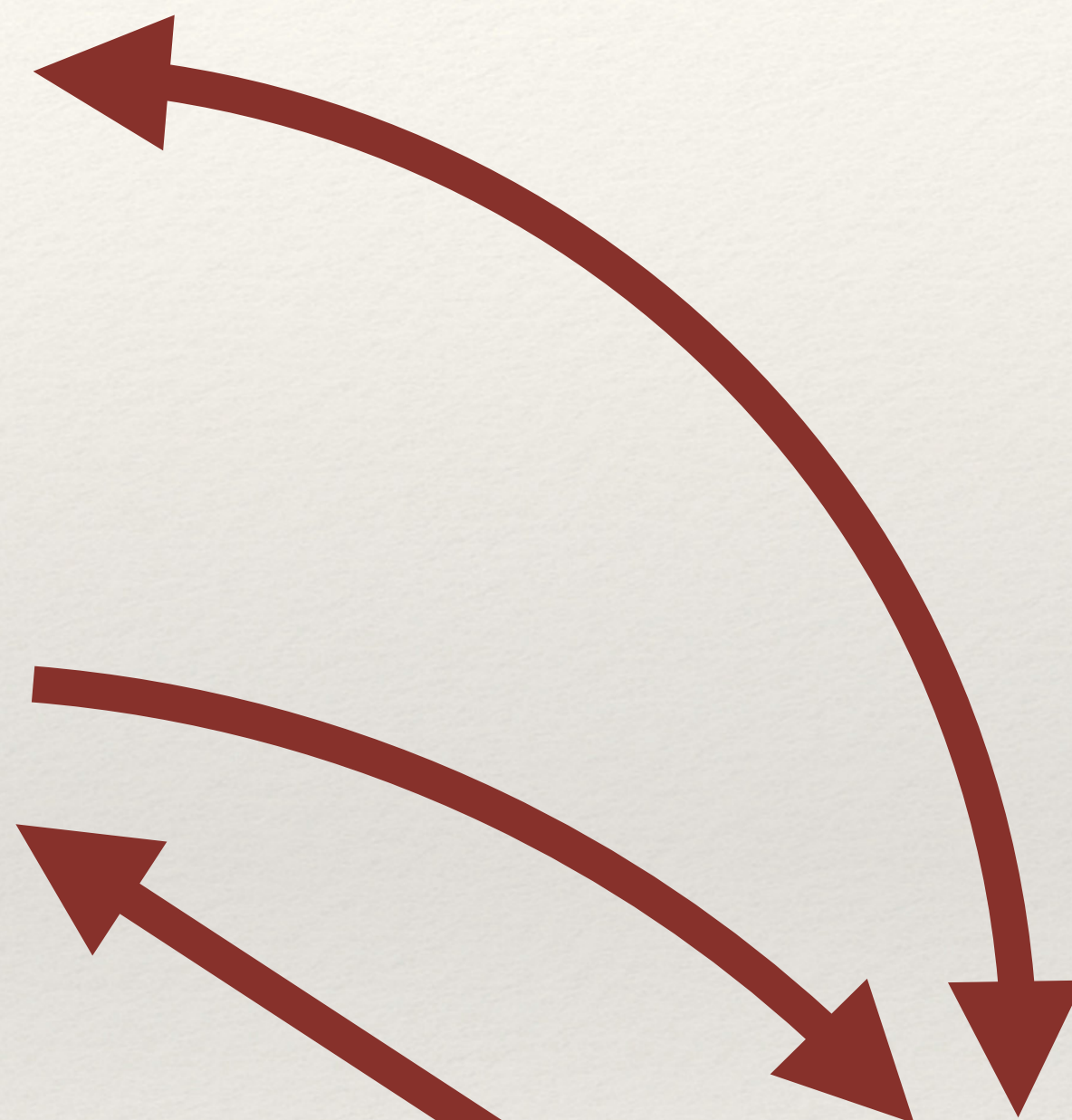
1st order

IRFs

0th order

Steady state

Endog. portfolios



What does second order perturbation give us?

Orders in aggregate risk

2nd order

Steady state
risk premium

1st order

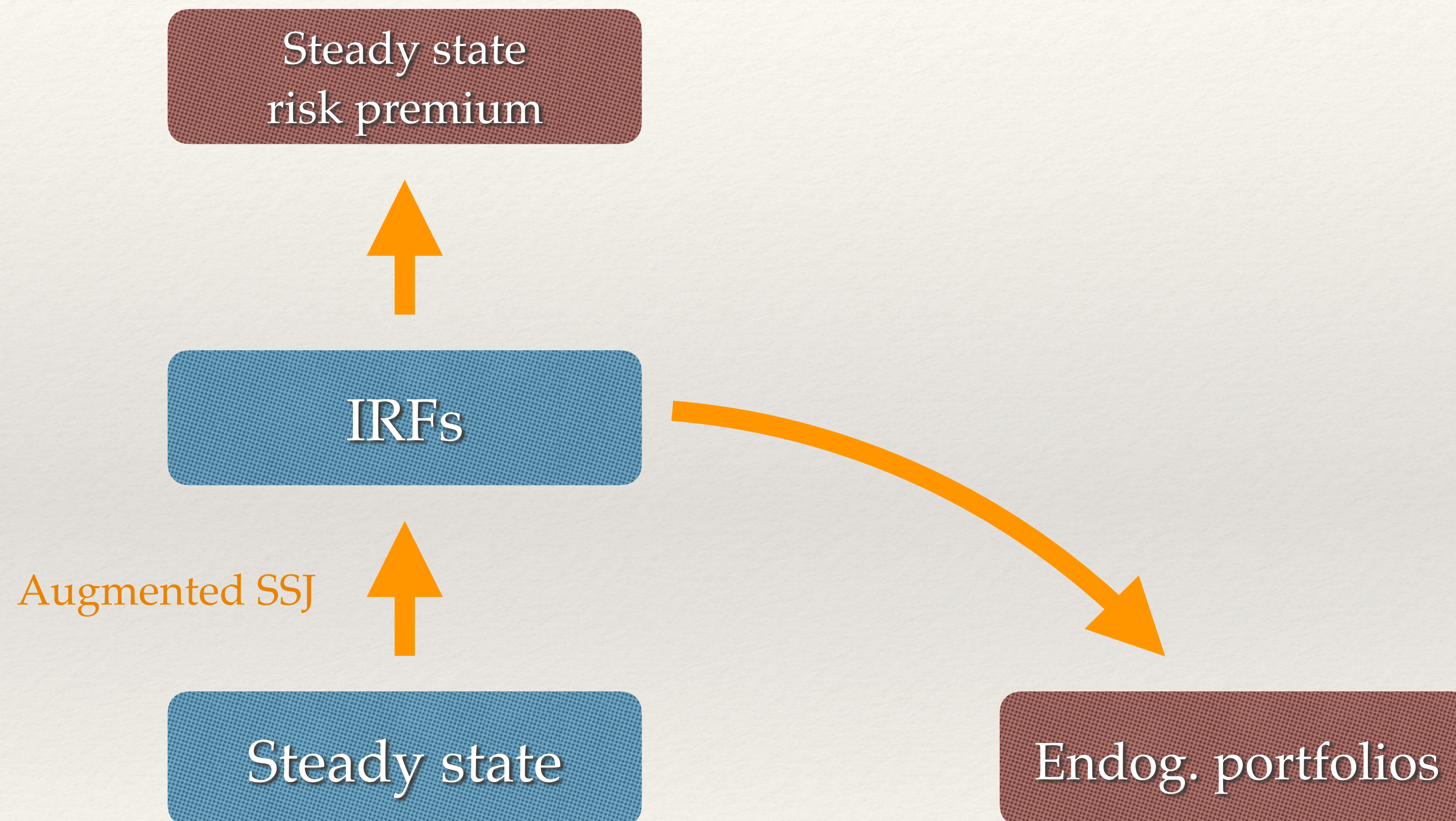
IRFs

Augmented SSJ

0th order

Steady state

Endog. portfolios



Canonical model with locally complete markets

General setting (one account, many assets)

- ❖ Heterogeneous households i can allocate wealth a_i to $K + 1$ assets
- ❖ Aggregate risk: random variable ϵ ; asset k has stochastic return $R^k(\epsilon)$
- ❖ Given value function V_i , problem of household i is:

$$\max_{\{a_i^k\}} \mathbb{E}_\epsilon \left[V_i \left(\sum_{k=0}^K R^k(\epsilon) a_i^k, \epsilon \right) \right] \text{ s.t. } \sum_{k=0}^K a_i^k = a_i$$

- ❖ Classic first-order condition, letting $\gamma_i \equiv$ multiplier on i 's budget constraint

$$\mathbb{E}_\epsilon \left[R^k(\epsilon) \frac{\partial V_i}{\partial a}(\epsilon) \right] = \gamma_i \quad \forall i, k$$

Implications of asset spanning

- ❖ Suppose finite ϵ 's (label one ϵ_{ss}). With enough assets, we get **complete markets**:

$$\frac{\partial V_i}{\partial a}(\epsilon) / \frac{\partial V_i}{\partial a}(\epsilon_{ss}) = \lambda(\epsilon) \quad \forall \epsilon$$

- ❖ λ is (cross-sectional) stochastic discount factor, gives us **risk premia**

$$\mathbb{E}_\epsilon [R^k(\epsilon) - R^0(\epsilon)] = -\text{Cov}_\epsilon \left(R^k(\epsilon) - R^0(\epsilon), \lambda(\epsilon) / \mathbb{E}_\epsilon [\lambda(\epsilon)] \right) \quad \forall k$$

- ❖ With continuous ϵ 's, but finite K , can still get this “locally” (ie to 1st-order)
 - ❖ 2nd-order perturbation to ϵ shows we need $K \geq \dim \epsilon + \text{spanning condition}$
 - ❖ “0th order portfolios” a_i^k ; well-defined in the limit as $\epsilon \rightarrow \epsilon_{ss}$

[eg Tille-van Wincoop 2010, Devereux-Sutherland 2011, Coeurdacier-Rey 2013, etc.]

Application to one-account, many-asset model

- ❖ Consider now standard one-account model with idiosyncratic risk e + aggregate risk

$$V_t(a_0, \dots, a_K, e) = \max_{c, a'_k} \frac{c^{1-\sigma}}{1-\sigma} + \beta \mathbb{E} \left[V_{t+1}(a'_0, \dots, a'_K, e') \right]$$

$$c + \sum_{k=0}^K a^{k'} = \sum_{k=0}^K R_t^k a^k + W_t e ; \quad \sum_{k=0}^K a^{k'} \geq \underline{a}$$

- ❖ Say all agg. risk realized at date 0. For $t \geq 1$: same return R_t , only $a \equiv \sum a_k$ matters

$$V_t(a, e) = \max_{c, a'} \frac{c^{1-\sigma}}{1-\sigma} + \beta \mathbb{E} \left[V_{t+1}(a', e') \right]$$

$$c + a' = R_t a + W_t e \quad a' \geq \underline{a}$$

Complete markets in the one-account model

- ❖ At $t = 0$, ϵ realizes, so paths $R_t^k(\epsilon)$ and $W_t(\epsilon)$ for $t \geq 0$ become known
- ❖ At $t = -1$, using envelope $\partial V / \partial a = c^{-\sigma}$, the risk-sharing condition is

$$\frac{\mathbb{E}_e \left[c_0^{-\sigma} \left(a_{ss}(t_0(a_{ss}, e_-), \epsilon) \right) \right] e_-}{\mathbb{E}_e \left[c_{ss}^{-\sigma} \left(a_{ss}(a_{ss}, e_-), \epsilon \right) \right] e_-} = \lambda(\epsilon) \lambda(\epsilon) \forall \epsilon \quad \forall \epsilon$$

- ❖ Can use this to **test** for portfolio optimality and **solve** for optimal portfolios
- ❖ Equivalently: solve for “transfers” $t_0(a_{ss}, e_-, \epsilon)$ relative to baseline portfolios
- ❖ Later, figure out what portfolios decentralize these transfers

First-order complete markets heuristic

- ❖ Write ∂c_{i0} for date-0 consumption response to shock under baseline portfolios
- ❖ To first-order, risk sharing condition says (heuristically):

$$\frac{dc_{i0}}{c_{i0}} = \frac{\partial c_{i0} + mpc_{i0}dt_{i0}}{c_{i0}} = \frac{-d\lambda}{\sigma}$$

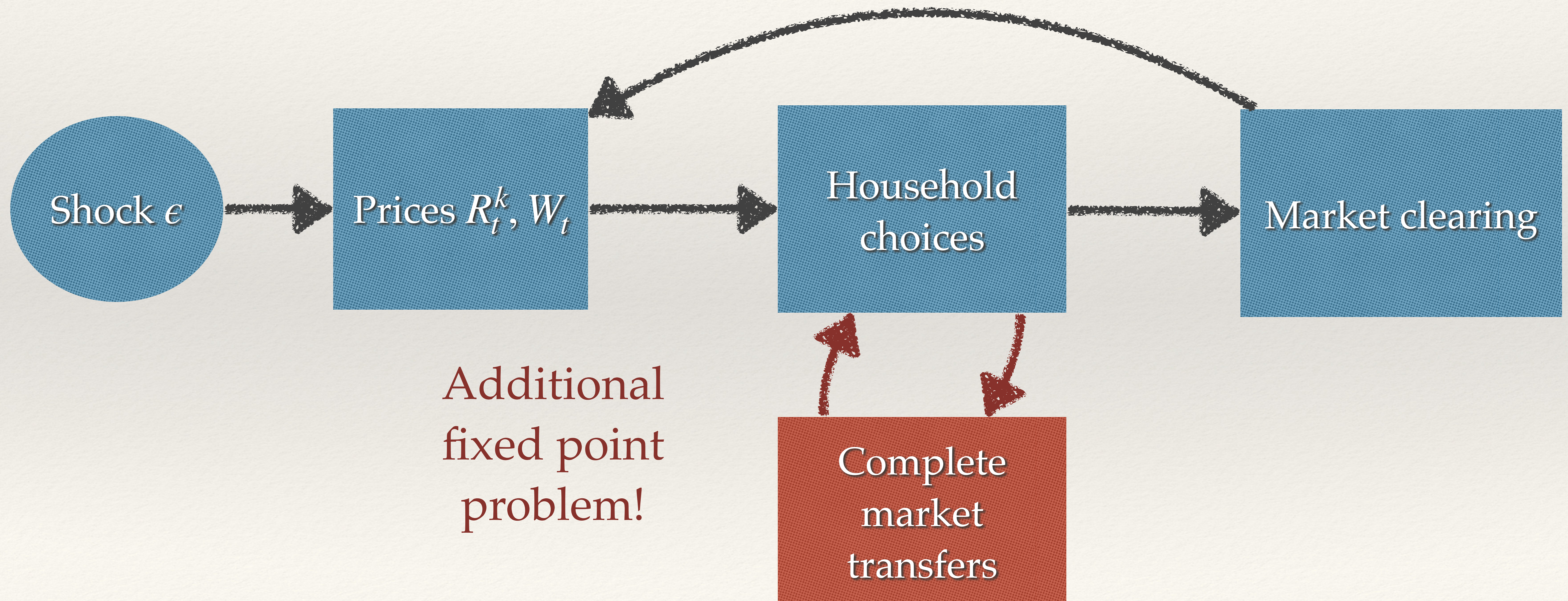
- ❖ Impose $\mathbb{E}_I[dt_{i0}] = 0$ to solve for sdf $d\lambda$ and equilibrium transfer dt_{i0} :

$$\frac{-d\lambda}{\sigma} = \frac{dc_{i0}}{c_{i0}} = \mathbb{E}_I \left[\frac{\frac{c_{i0}}{mpc_{i0}}}{\mathbb{E}_I \left[\frac{c_{i0}}{mpc_{i0}} \right]} \frac{\partial c_{i0}}{c_{i0}} \right] \quad dt_{i0} = \frac{\frac{-d\lambda}{\sigma} c_{i0} - \partial c_{i0}}{mpc_{i0}}$$

➡ Complete markets ~undo the heterogeneous consumption effects of shocks

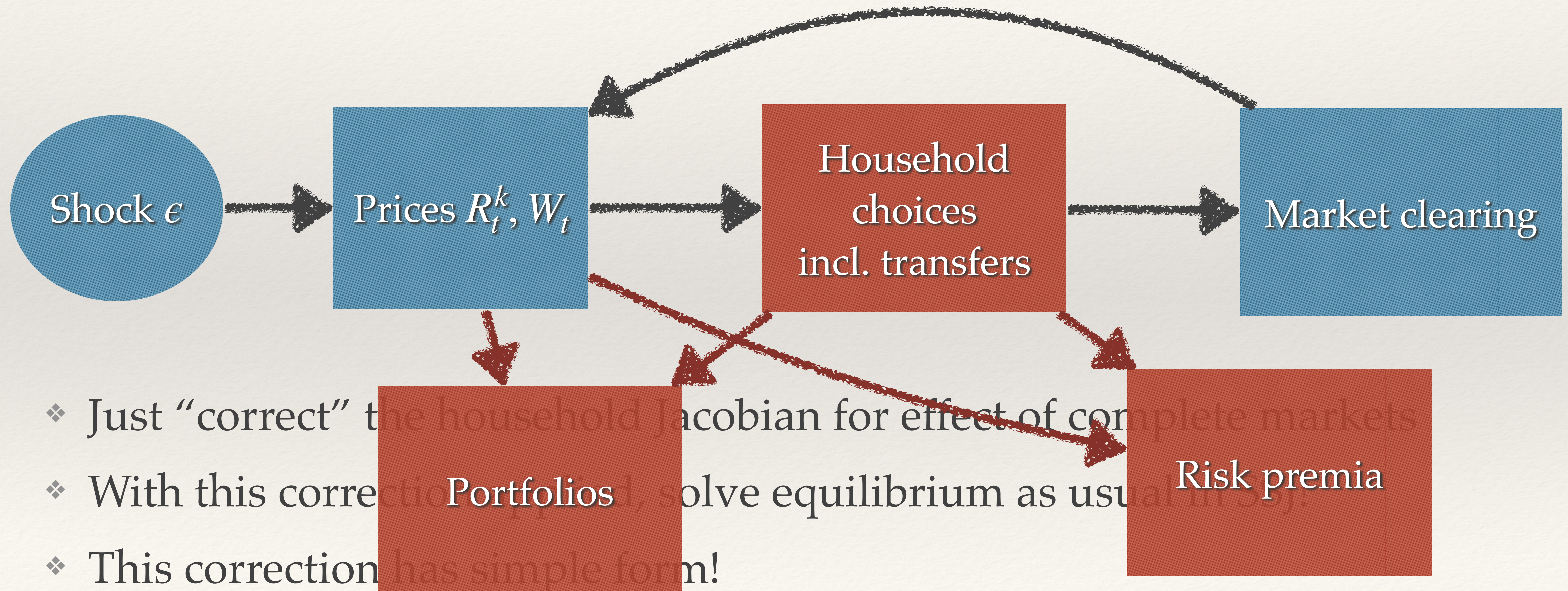
General equilibrium

- ❖ Consider now general equilibrium:



General equilibrium solution with SSJ

- ❖ Tackle this with SSJ as follows



- ❖ Just “correct” the household Jacobian for effect of complete markets
- ❖ With this correction, solve equilibrium as usual
- ❖ This correction has simple form!

Implementation of correction in SSJ

- ❖ Intertemporal MPC matrix $\mathbf{M}$: $M_{ts} = \partial C_t / \partial W_s$
- ❖ Wage change at s has direct effect on policies ∂c_{0i}^s
- ❖ From optimal risk-sharing, get date-0 transfer dt_{0i}^s
- ❖ Together, these imply change in distribution of assets coming into date 0 $\mathcal{D}_s$
- ❖ In turn, this affects consumption at t by $\mathcal{E}_t' \mathcal{D}_s$ ($\mathcal{E}_t$ is expectation function for C)
- ❖ Hence, **simple correction** to sequence space Jacobian gives us effect under CM:

$$M_{t,s}^{CM} \equiv M_{t,s}^{exo} + \mathcal{E}_t' \mathcal{D}_s$$

- ❖ See notebook for additional details!

Sanity check: replicates known state-space results!

- ❖ Consider original Krusell-Smith (97) model: TFP shock, 2 assets capital+bond
- ❖ Portfolio constraints $k \geq 0, b \geq \underline{b}$ [needs modified, iterative algorithm, cf paper]

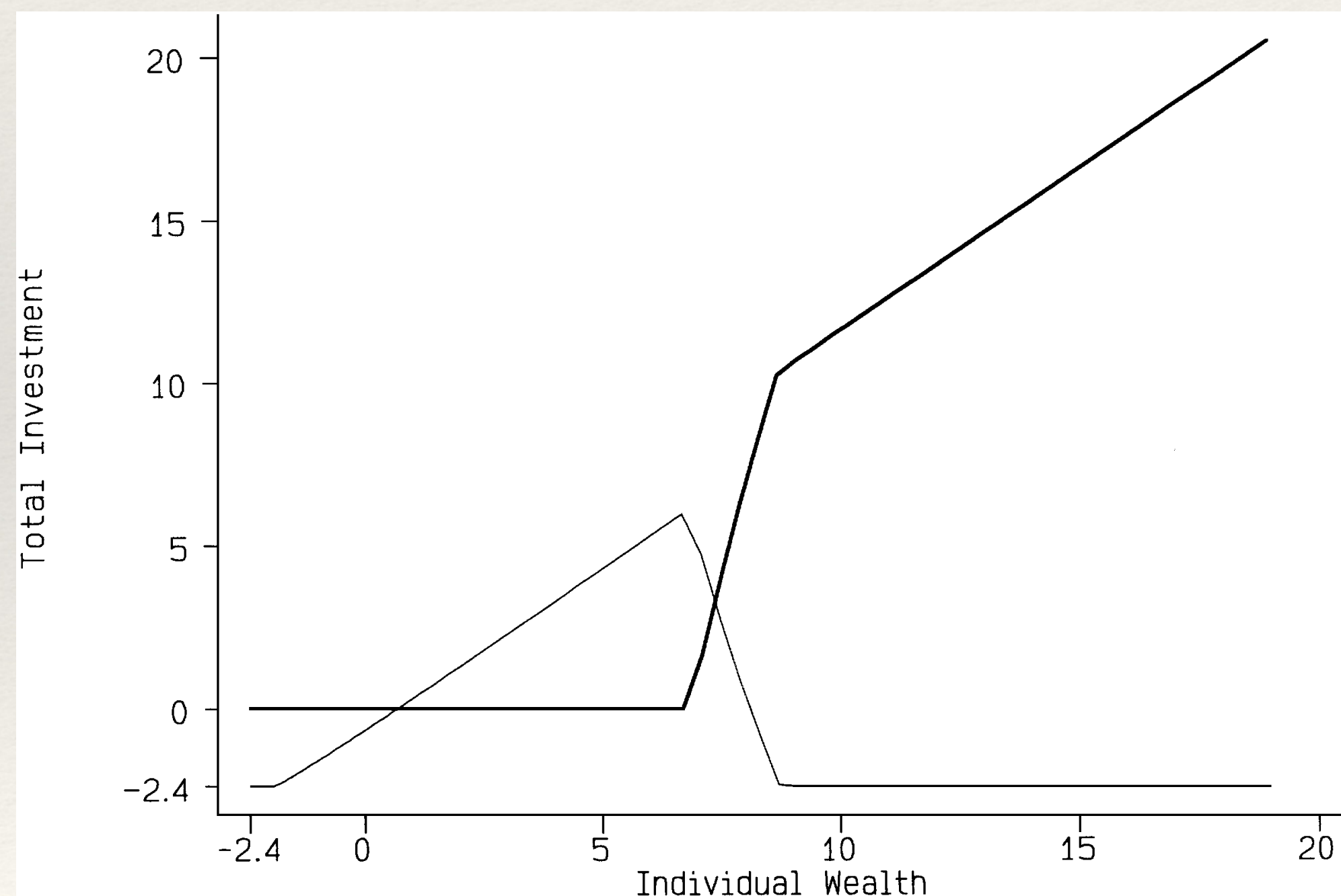
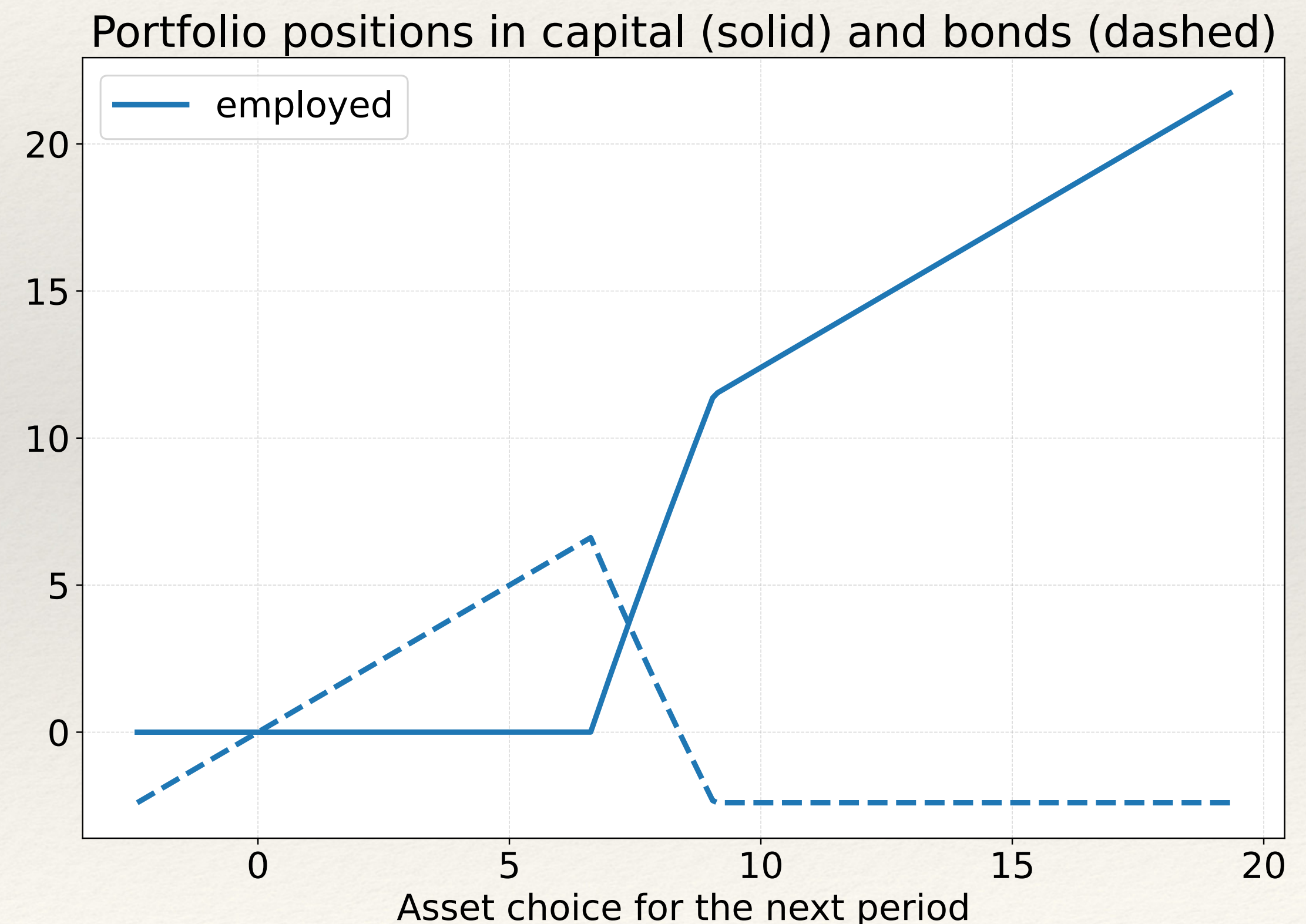


FIGURE 3. Portfolio decision rules of an employed agent: thick line = capital, thin line = bonds



Application to canonical HANK model with traded stocks

Canonical HANK model

- ❖ Application to canonical HANK model
- ❖ We now let households trade stocks and bonds directly:

$$\max_{c_{it}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v(N_t) \right)$$

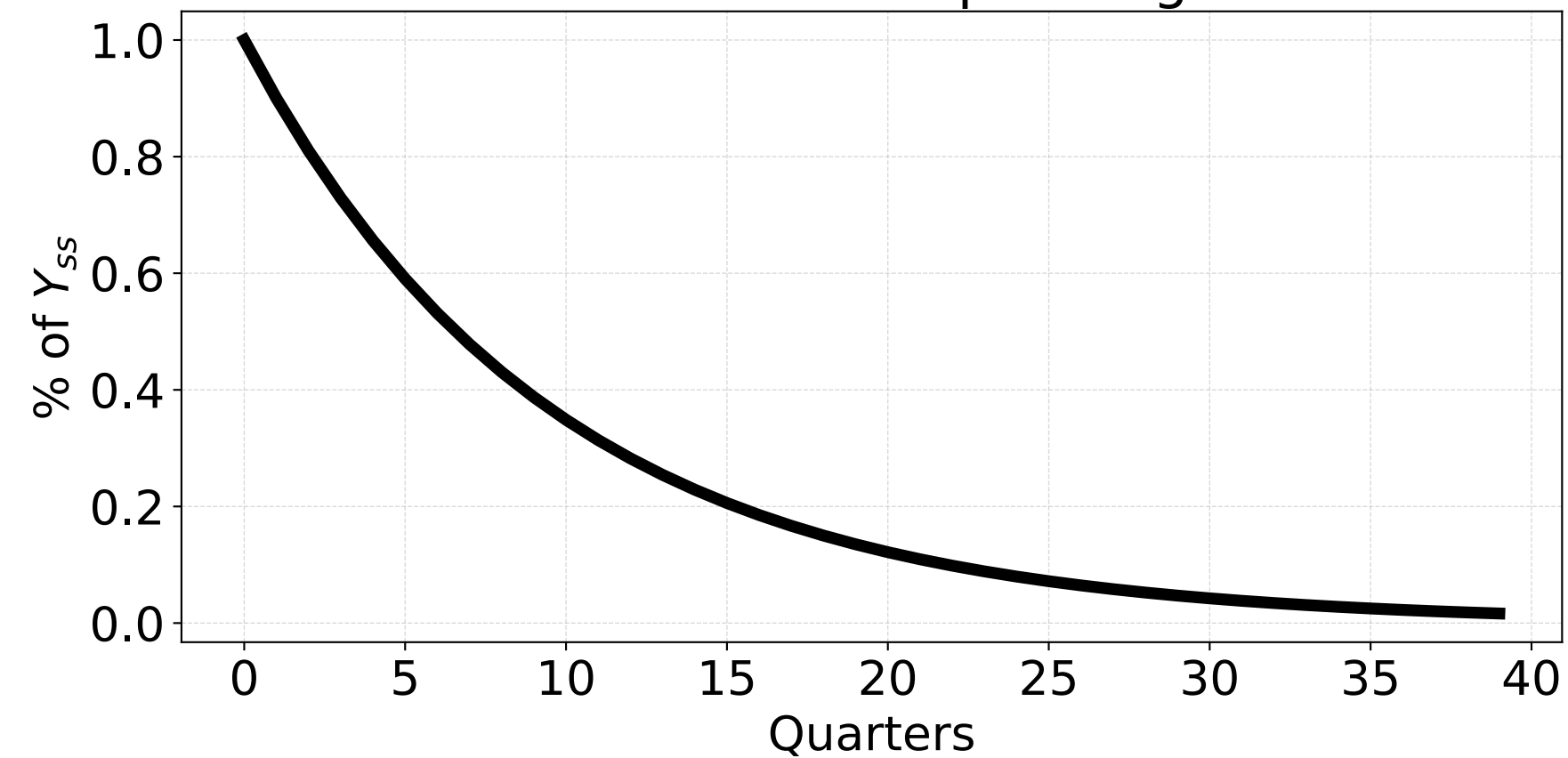
$$c_{it} + p_t s_{it} + b_{it} \leq (p_t + d_t) s_{it-1} + (1 + r_{t-1}) b_{it-1} + (1 - \tau_t) \frac{W_t}{P_t} e_{it}$$

$$p_t s_{it} + b_{it} \geq 0$$

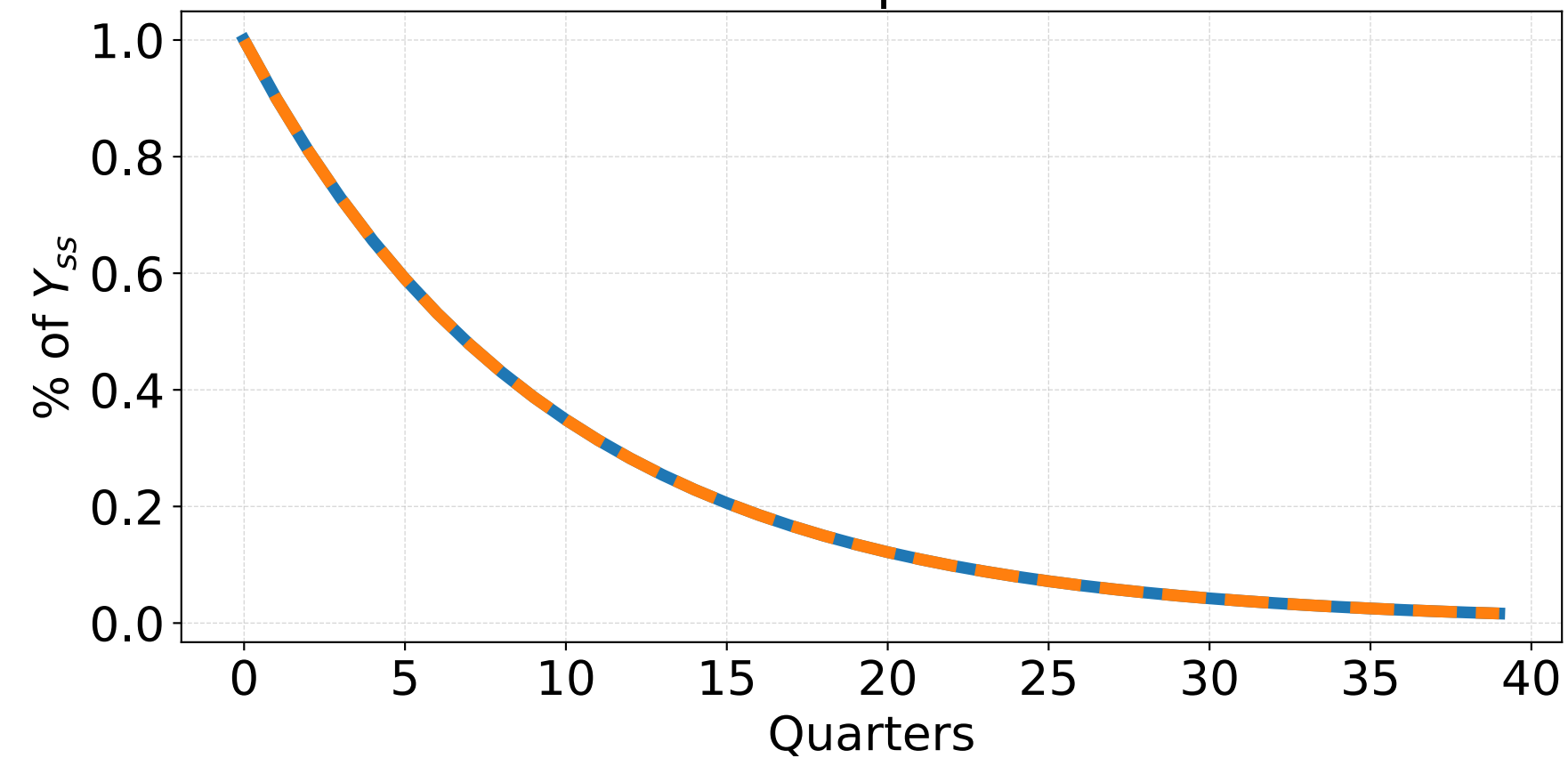
- ❖ s stocks (price p_t , dividends d_t), b bonds, assume $\sigma = 1$
- ❖ Rest of the model is as usual, $B_{ss} = 0$ (so Werning equivalence result applies)

Example 1: balanced-budget $\{G_t\}$ shock

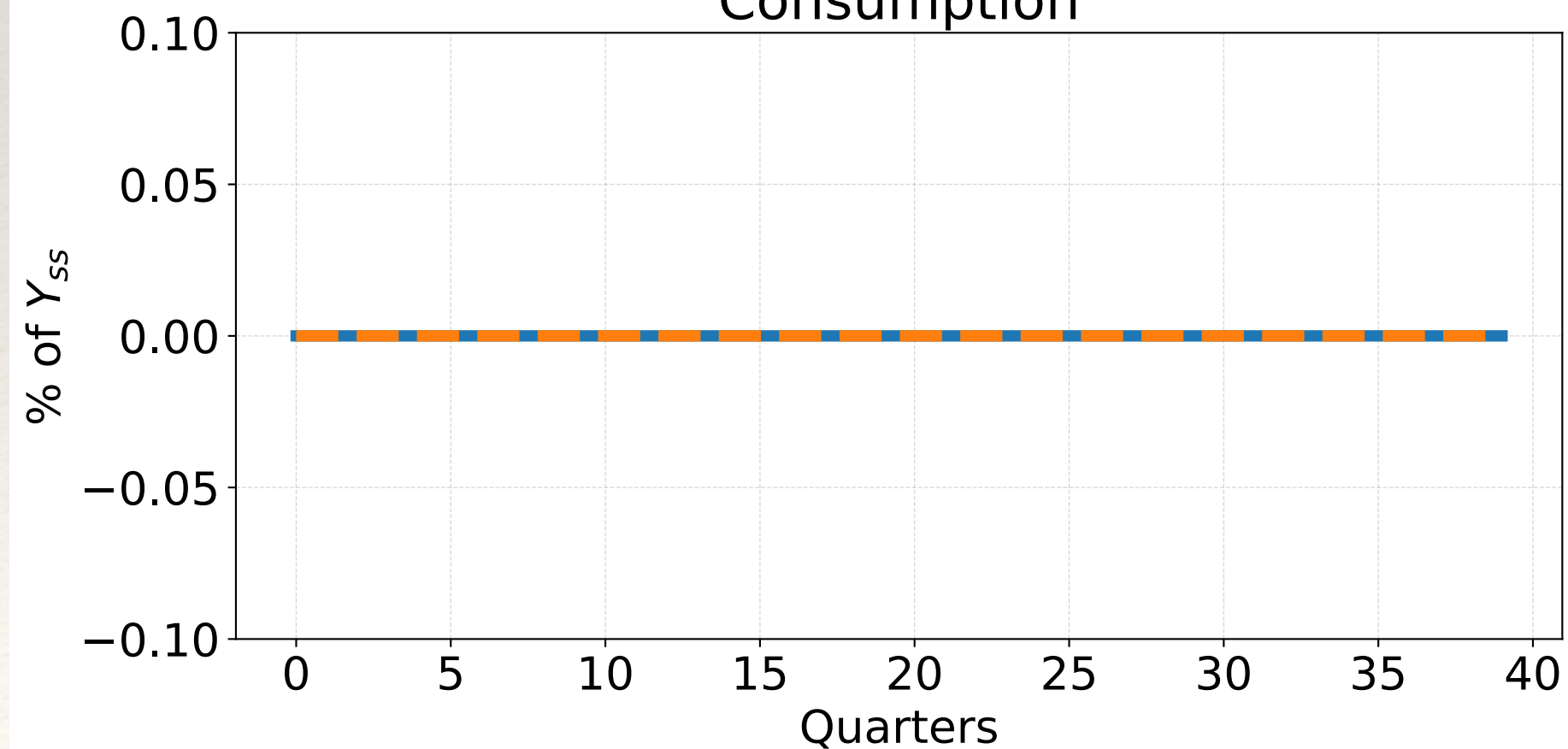
Government spending



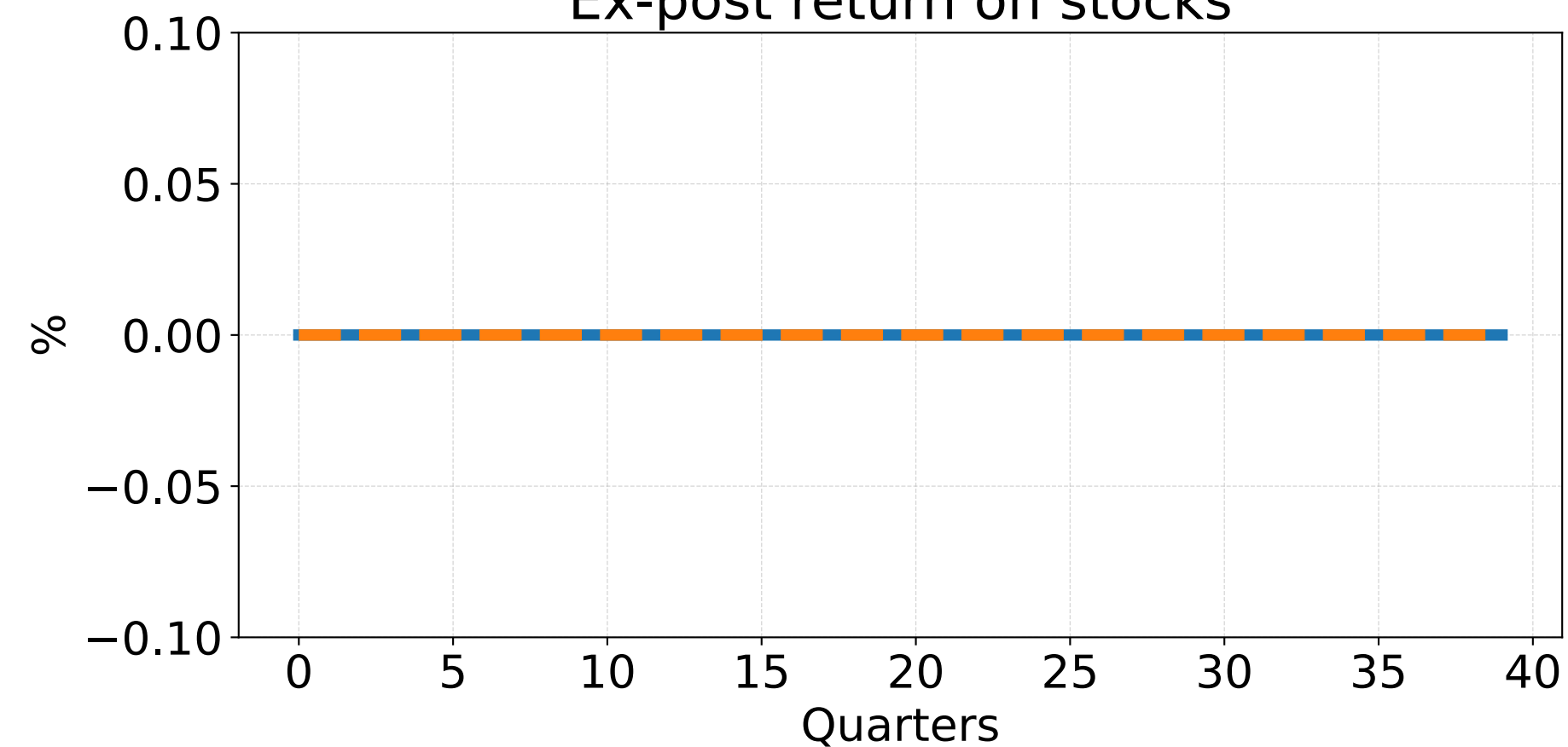
Output



Consumption



Ex-post return on stocks



— Exogenous portfolios (100% in stock market) - - Endogenous portfolios

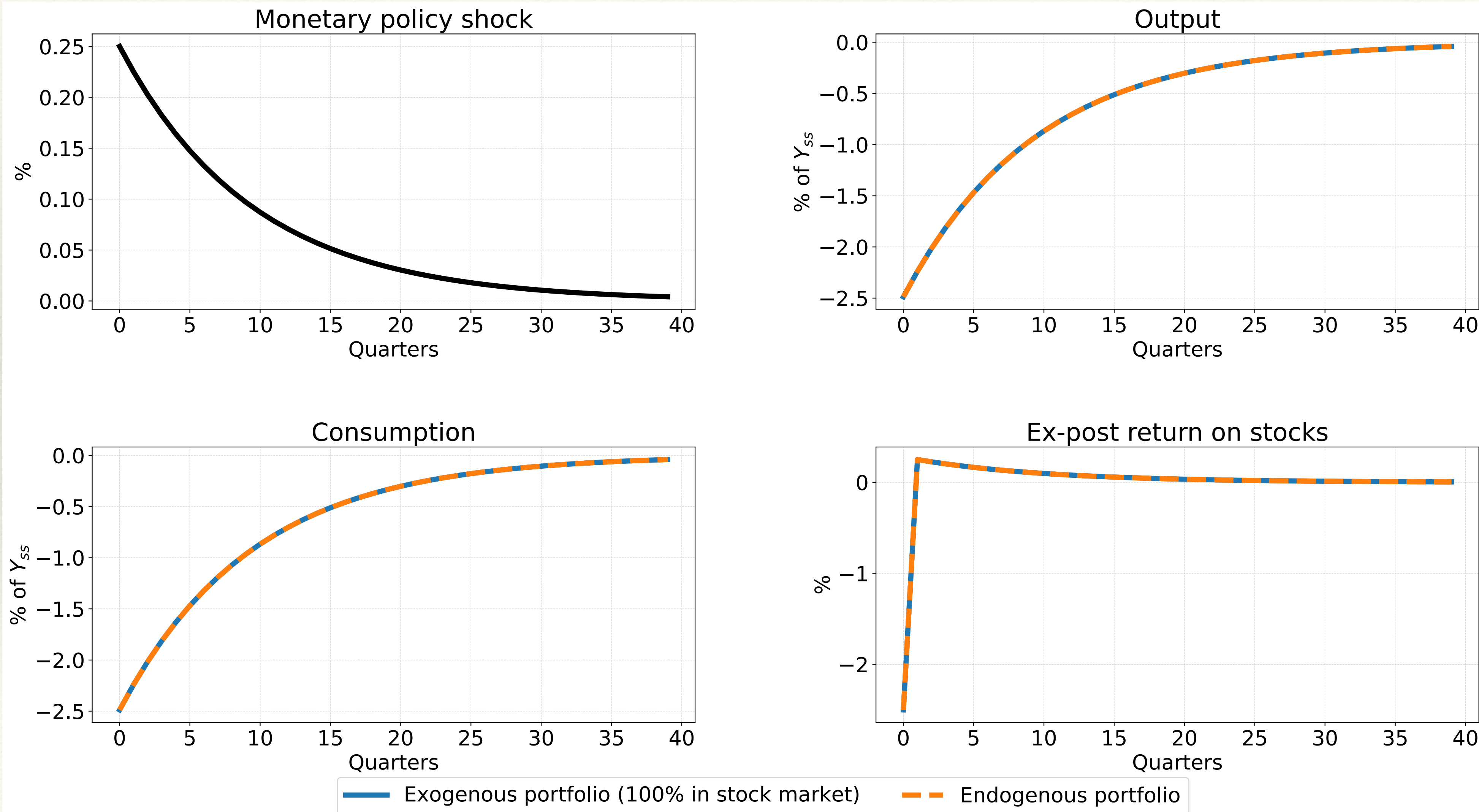
No effect from
portfolio choice!

Why? Homogeneous
consumption effects at
exogenous portfolios,
 $dc_{i0} = 0$

For endog. portfolios
to matter, need
unequal consumption
effects of shocks

Also, here stock prices
constant $\rightarrow$ no
difference between
bonds and stocks

Example 2: monetary policy $\{r_t\}$ shock



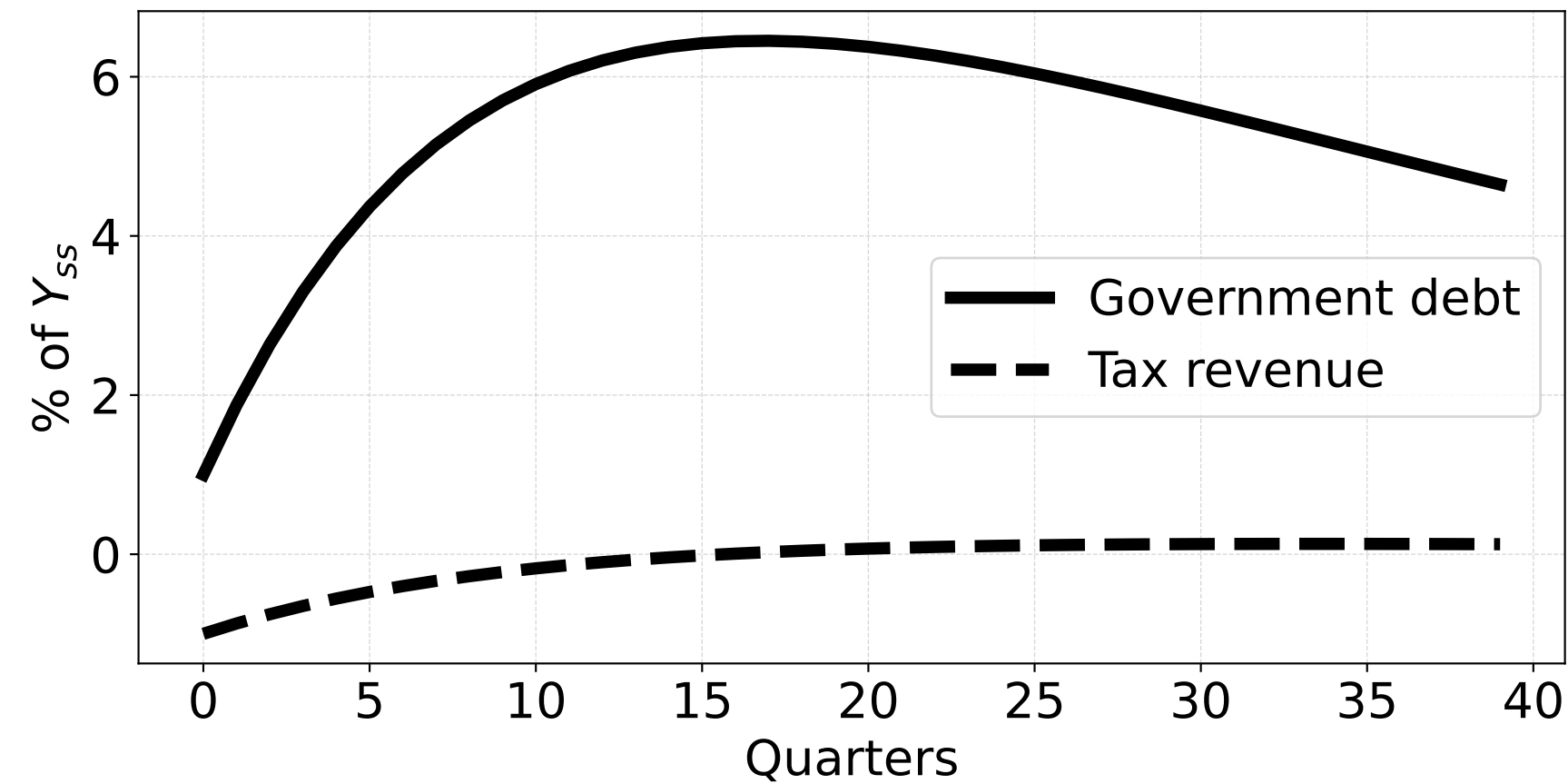
Still no effect from
portfolio choice!

100% stock portfolios
+ log utility:
homogeneous effects
of monetary policy
(Werning result!)

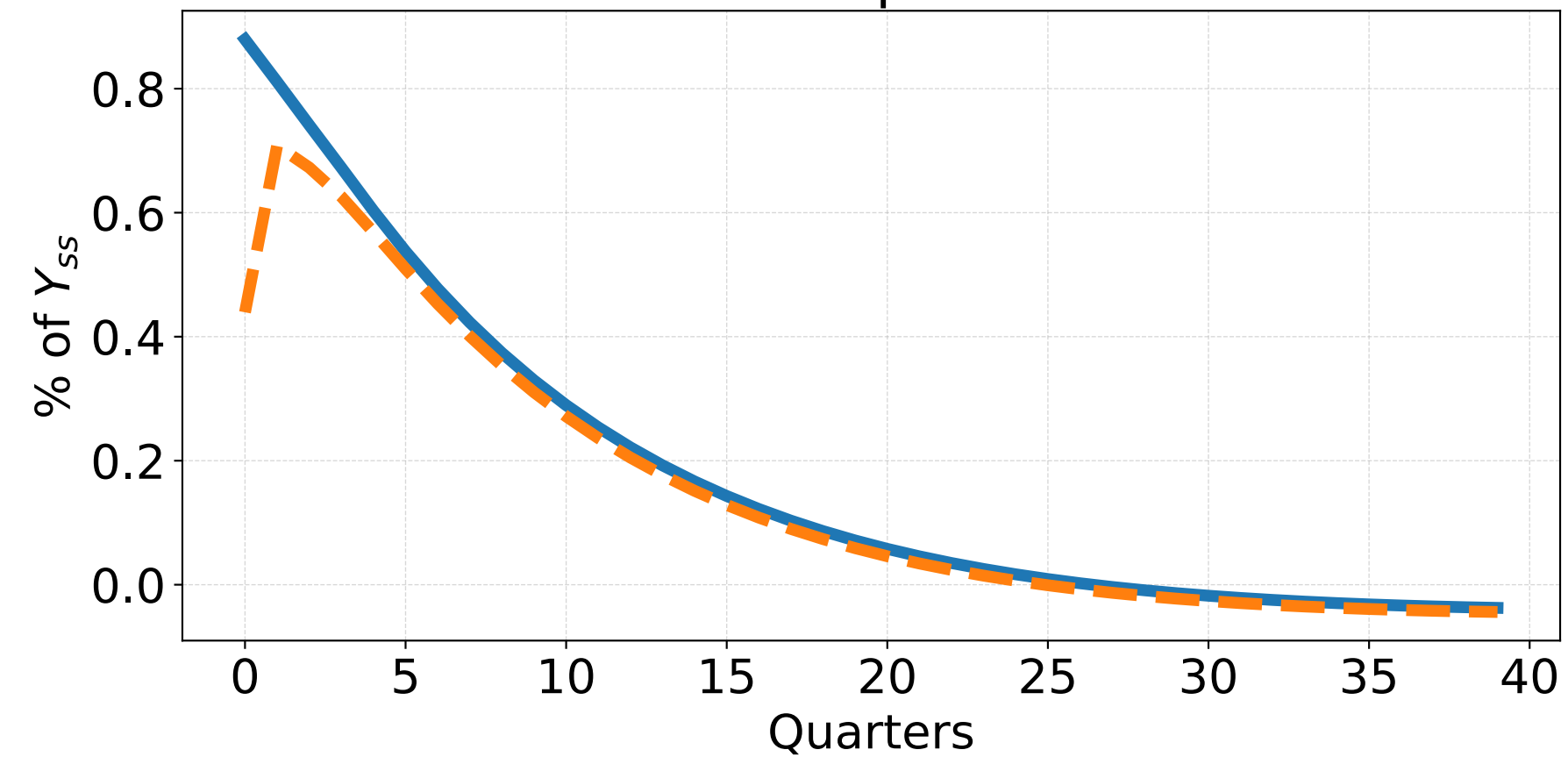
100% stocks are only
optimal portfolios here

Example 3: deficit-financed transfer $\{B_t\}$ shock

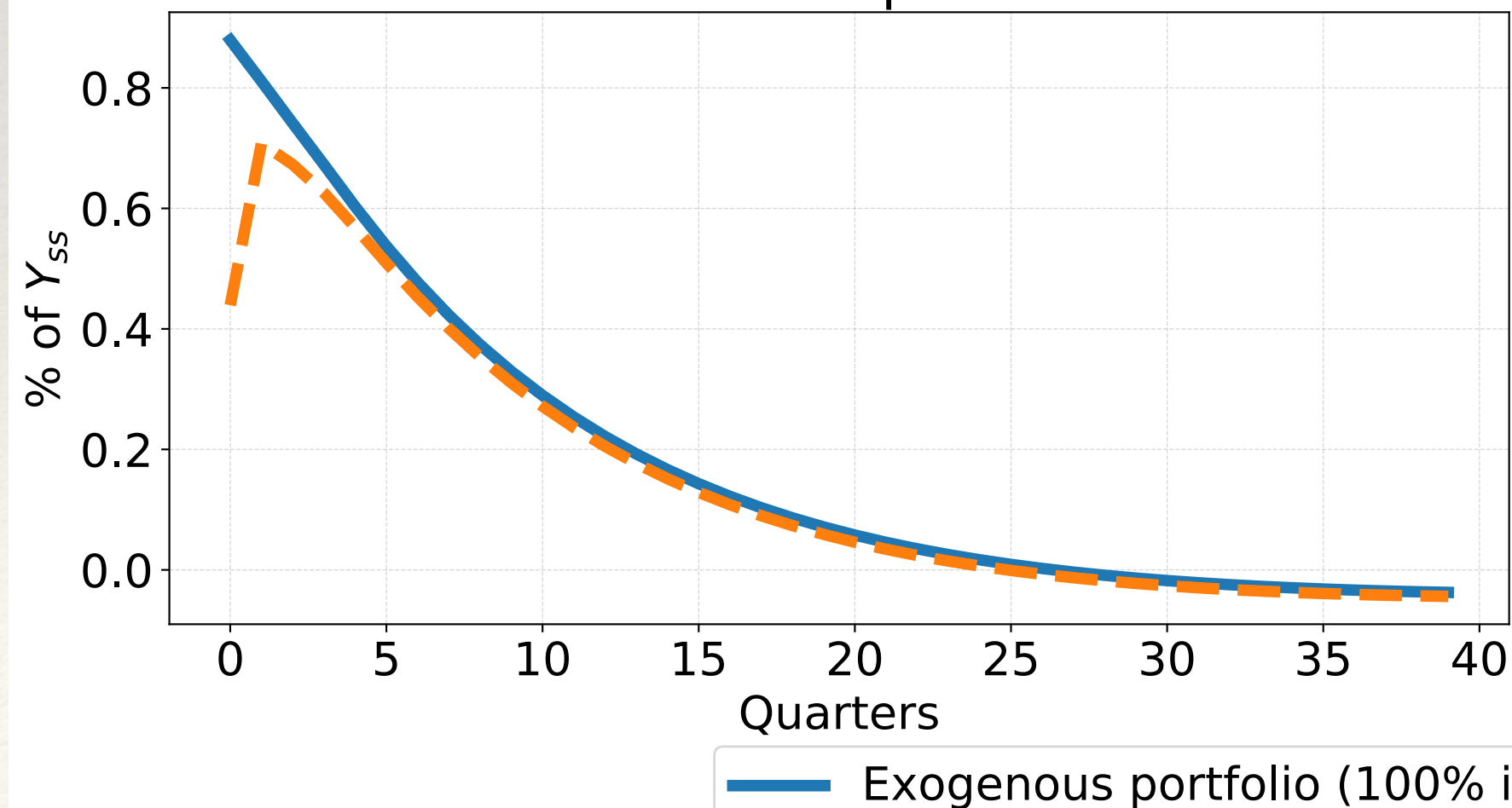
Deficit-financed shock



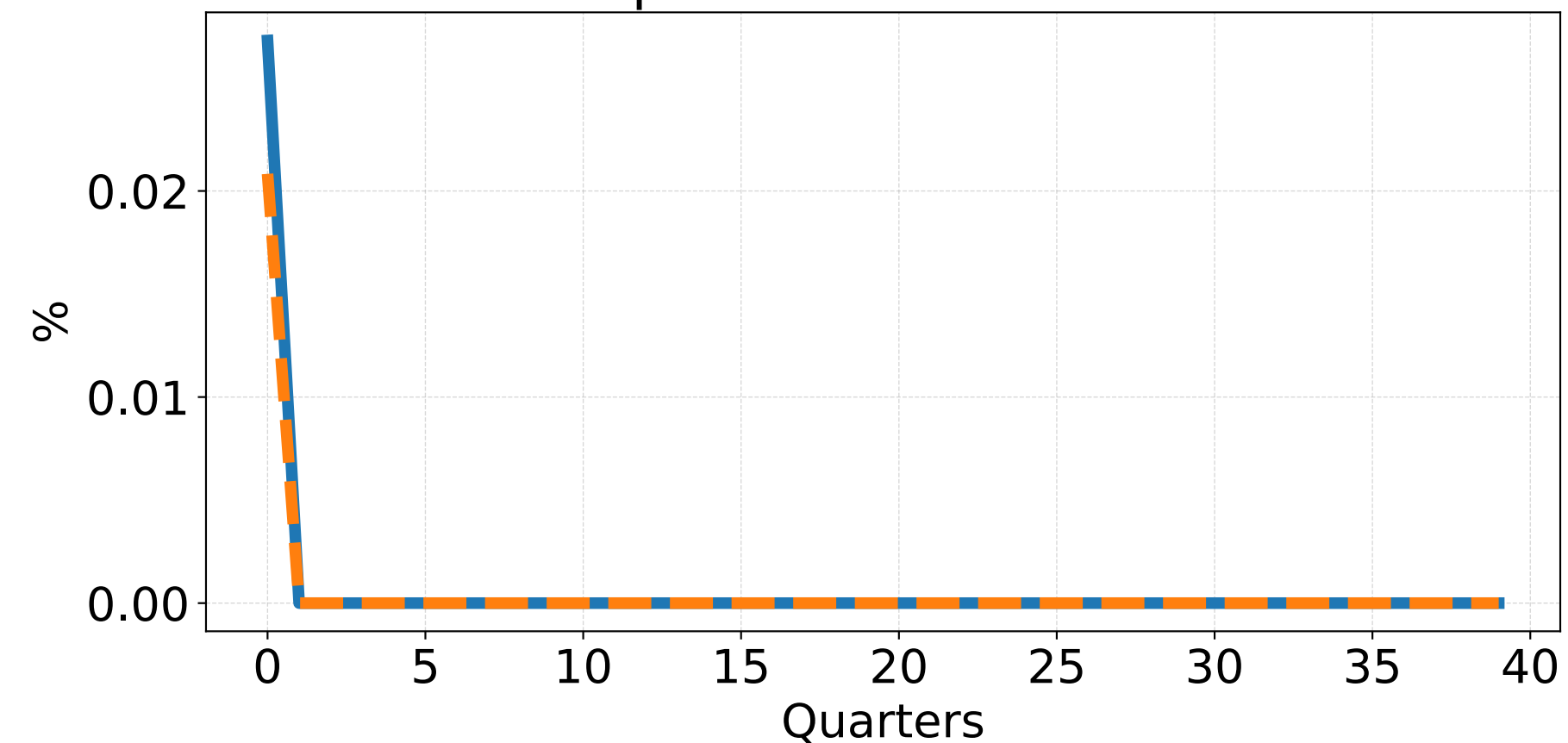
Output



Consumption



Ex-post return on stocks



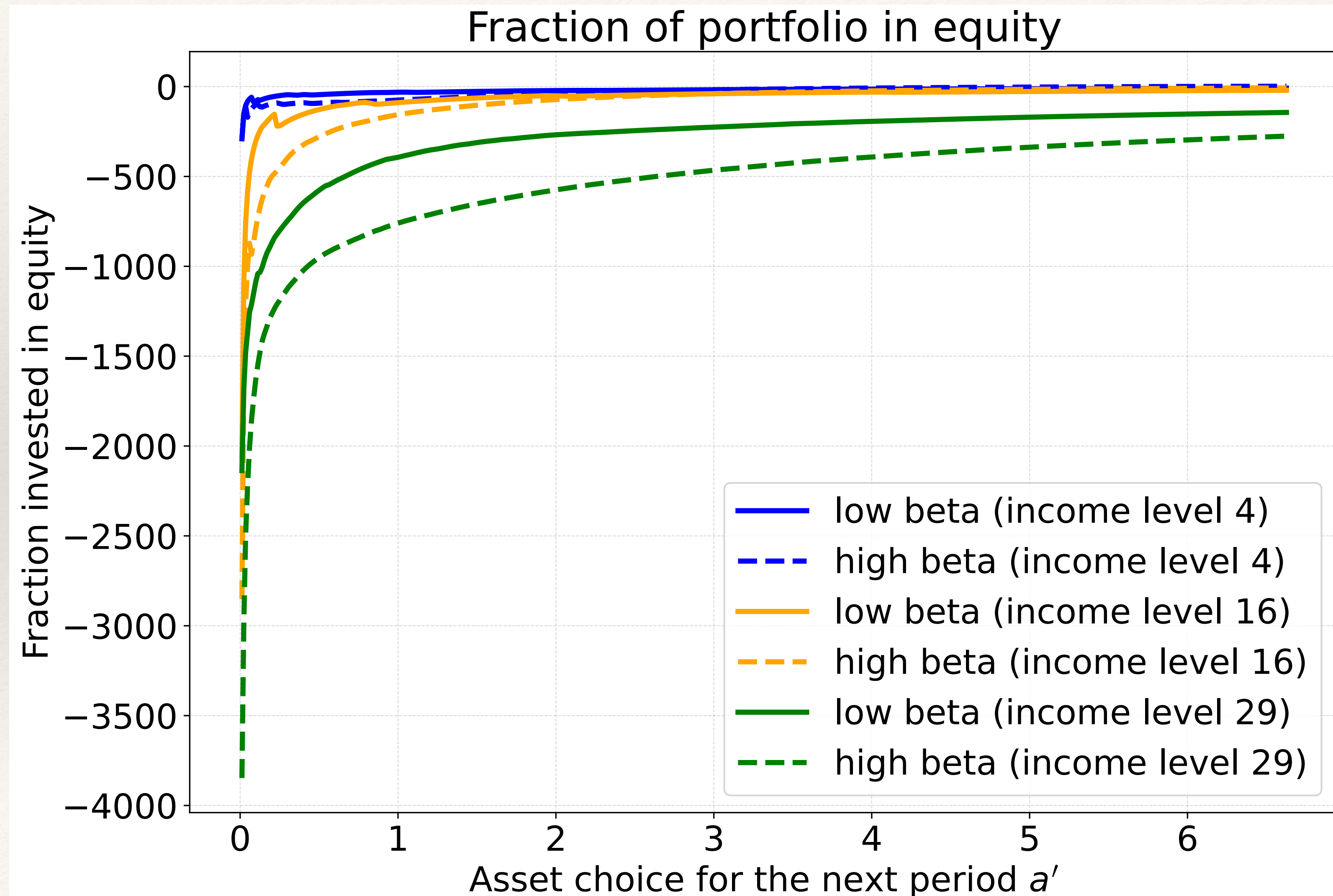
Large baseline output effects of deficit financed fiscal policy....

Largely undone by endogenous portfolios!

Poor disproportionately affected by shock, reduce stock exposure

But... how?

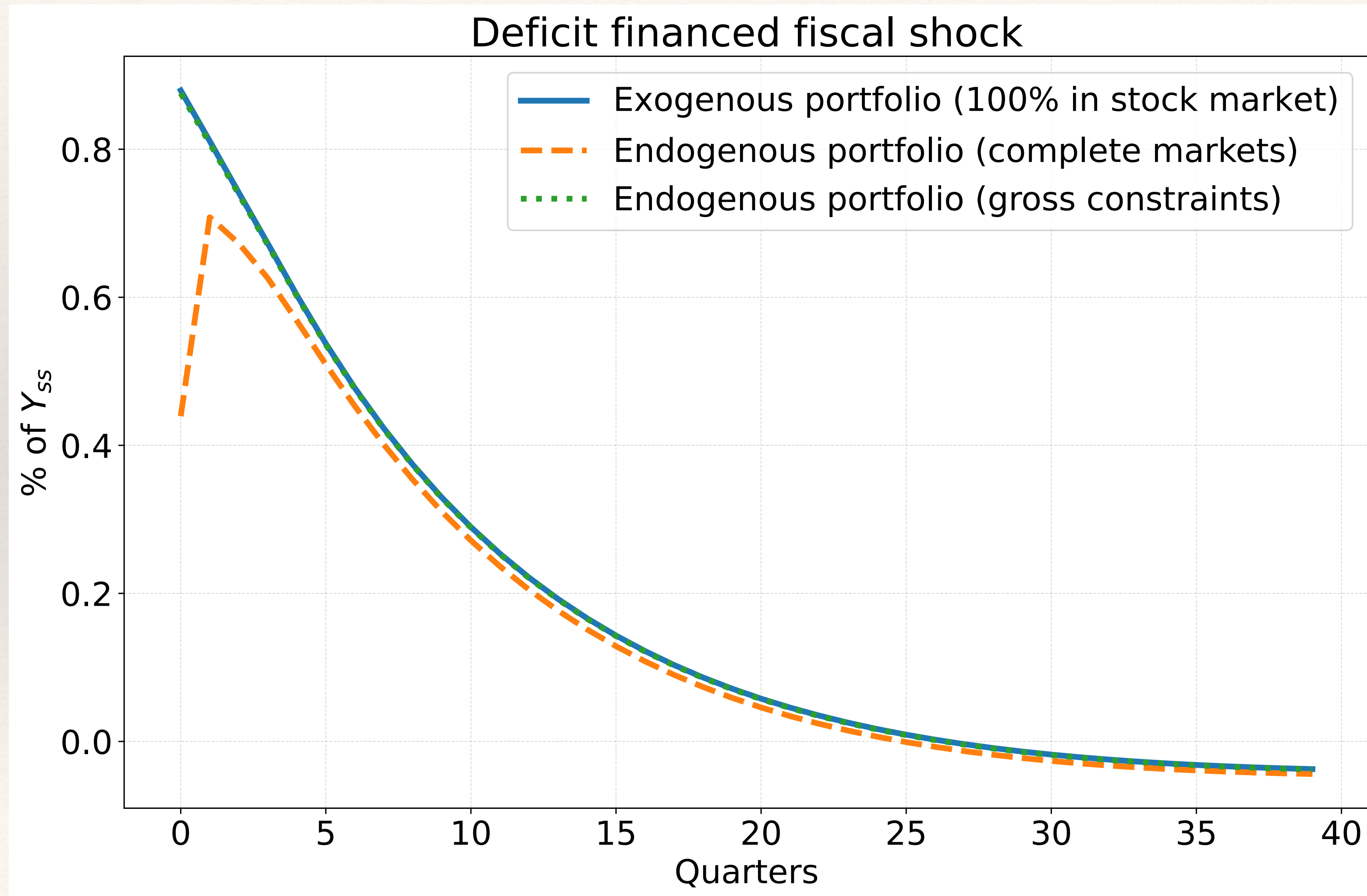
Under the hood: implausible portfolio shares!



We didn't restrict gross positions in assets: borrowing constraint applied only to net position!

So “complete markets” transfers achieved with ultra-levered short-selling by the poor.

With portfolio constraints... (no short sales, 1.5x leverage limit)



Constrained endogenous portfolio now ~same as exogenous portfolio: no effect!

For endog. portfolios to matter, need high-MPC agents to be able to take large gross positions

Conclusion

- ❖ Simple modification of sequence-space Jacobian algorithm gives us:
 - ❖ impulses with endogenous portfolios and second-order risk premia
 - ❖ can add portfolio constraints, incomplete markets (see paper)
 - ❖ new directions for asset pricing + heterogeneous agent macro
 - ❖ new directions for sequence-space macro
- ❖ When do endogenous portfolios matter in canonical HANK model?
 - ❖ When shocks have heterogeneous consumption effects
 - ❖ When high MPC agents are able to take highly leveraged positions